



Lesson 3: Move It, Look It, Sound It!

Big Idea

Motion, Looks, and Sound blocks are the three basic superpowers coders use to bring any sprite to life.

Learning Objectives

- Use motion blocks (move, turn, go to x/y, glide-like changes) to move a sprite around the stage.
- Use looks blocks (say, think, switch costume, change size, show/hide) to change how a sprite appears.
- Use sound blocks (play sound, play sound until done, change volume) to add sound effects.
- Combine all three block types into one working script.
- Understand the x/y coordinate grid as "an address" for where a sprite stands on stage.

Teacher Preparation

- Review x/y coordinates: x is left/right (negative = left, positive = right), y is up/down (negative = down, positive = up), center of stage is (0,0).
- Prepare a large printed or drawn grid (or draw one on the whiteboard) to demonstrate x/y visually before going on the computer.
- Remind students: "Remember Lesson 2, when we made our sprite move and talk for the first time? Today we go MUCH further with all three superpowers together."

Required Files

- [Student_Lesson_Guide.md](#)
- [Activities.md](#)
- [Challenges.md](#)

Teaching Flow (approx. 45 minutes)

1. Warm-up / Hook — 5 min
2. Concept Introduction — 10 min

3. Teacher Demonstration — 8 min
4. Guided Coding — 10 min
5. Mini Challenge — 5 min
6. Main Activity — 5 min
7. Check for Understanding + Recap — 2 min

Warm-up / Hook (5 min)

 **Coding Mission:** Play a quick "Simon Says" style game: "Simon says take one step right! Simon says turn around! Simon says freeze and shout 'Hooray!'" Connect it: "That's exactly what motion, looks, and sound blocks do for our sprites — just like your body just moved, turned, and made a sound!"

Concept Introduction (10 min)

- **Motion blocks (blue):** move the sprite around — think of them as "sprite feet." Introduce the x/y grid: "Imagine the stage is a giant map. x tells you left or right, y tells you up or down. The very middle of the stage is (0,0)."
- **Looks blocks (purple):** change how the sprite appears or what it says — think of them as "sprite face and clothes." Costumes are like different outfits or poses.
- **Sound blocks (pink/magenta):** make the sprite make noise — think of them as "sprite voice."
-  **Think Like a Coder:** "A sprite with only motion blocks can move, but it can't tell you how it feels. A sprite with only looks blocks can talk, but it can't go anywhere. Real characters need ALL of these together — just like you use your legs, face, and voice together every day!"

Teacher Demonstration (8 min)

1. Show "move 10 steps" vs "go to x: 0 y: 0" — explain the difference (move is relative/step-by-step, goto is an exact teleport to a spot).
2. Show "turn right 15 degrees" a few times in a row, and how the sprite spins.
3. Show "switch costume to [costume2]" and "next costume," pointing out the Costumes tab where new looks are drawn/chosen.
4. Show "change size by 10" a few times, then "set size to 100" to reset.
5. Show "play sound [meow] until done" — explain the difference between "play sound" (keeps going) and "play sound until done" (waits for it to finish before the next block runs).

Guided Coding (10 min)

Students build, step by step, alongside the teacher: 1. "go to x: -100 y: 0" (start position, like home base) 2. "move 10 steps" repeated by clicking, OR a single "move 50 steps" 3. "next costume" 4. "play sound [pop] until done" 5. "say [Wheee!] for 2 seconds" Run the full stack and watch it all happen together.

Mini Challenge (5 min)

Rearrange the ORDER of your blocks (say first, then move, then sound) and predict what changes before running it. Discuss: does order matter? (Yes! Code runs top to bottom.)

Main Activity / Project (5 min)

"Sprite Show": Students create a short 4–6 block sequence combining at least one motion block, one looks block, and one sound block, telling a tiny "moment" (like a sprite waking up, stretching, and yawning with a sound).

Fun Challenge (optional, harder) ★

Use "glide"-style repeated small moves plus a costume change on each step to make your sprite look like it's really walking (move a little, switch costume, move a little, switch costume again).

Check for Understanding

- "What's the difference between 'move' and 'go to x/y'?"
- "What block would you use to make a sprite look surprised or happy?"
- "What happens if you use 'play sound' instead of 'play sound until done' right before a 'say' block?"
- "Does the order of blocks in a stack matter? Why?"

Common Student Mistakes (with fixes)

- **Mistake:** Confusing "move" (relative) with "go to x/y" (absolute position). **Fix:** Show side by side: click move 3 times in a row (sprite keeps going) vs go-to-x-y 3 times with the same numbers (sprite doesn't move after the first click, because it's already there!).
- **Mistake:** Sprite "disappears" off the edge of the stage. **Fix:** Use "go to x:0 y:0" to bring it back, and discuss the stage boundaries (roughly x: -240 to 240, y: -180 to 180).
- **Mistake:** Using "play sound" (not "until done") before a say block, so the say block runs immediately and cuts the sound off in perception. **Fix:** Show them "play sound until done" so blocks below wait their turn.

-  **Debug Detective:** "A student's sprite is supposed to grow bigger every time it's clicked, but it just keeps getting bigger and bigger forever, off the screen! What happened?" (Answer: they used "change size by" repeatedly without ever resetting with "set size to 100" — walk through adding a reset step.)

Differentiation

- **Struggling support:** Provide a pre-built partial stack (move + say) and have them add just ONE sound block.
- **Standard:** Full guided + main activity as planned.
- **Extension for advanced learners:** Challenge them to make a sprite "walk in a square" using four move + turn combinations (move, turn right 90, move, turn right 90, move, turn right 90, move, turn right 90) and predict the shape before running it.

Recap

"Today, your sprites got THREE new superpowers: moving with motion blocks, changing looks with looks blocks, and adding sound with sound blocks. You even discovered that the ORDER of blocks changes what happens!"

Take-Home / Creative Challenge

At home (no computer needed), act out a tiny 3-part "show" using your own body: one movement, one facial expression or costume change (like putting on a hat), and one sound effect you make with your voice. Perform it for a family member!

Teacher Talking Points

- "You just made your sprite do three things at once — moving, looking different, AND making sound. That's a real animated character!"
- "It's totally fine if your sprite flies off the stage — that just means we learned something about the x/y map together."
- "I noticed some of you are already predicting what a block will do BEFORE clicking it. That's exactly how real coders think!"



Victory Moment

Have a few volunteers show their "Sprite Show" to the class. Celebrate every single one: "Look at that — three superpowers, one awesome sprite moment!"